

- Q.30 What are the different types of Buses used by Embedded Systems?
- Q.31 What are the factors that affect the performance of an embedded system?
- Q.32 Discuss the steps involved in interfacing of 7-segment display.
- Q.33 What are the applications of Embedded Systems?
- Q.34 Explain the term simulator.
- Q.35 What are the challenges of embedded Software Development?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the embedded system architecture with diagram.
- Q.37 What is PIC microcontroller? Draw the block diagram and explain each block?
- Q.38 Draw the block diagram of AVR microcontroller and explain each block in detail.

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6th Sem / Eltx Subject:- Embedded Systems

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What does ISR stand for?
a) interrupt standard routine
b) interrupt service routine
c) interrupt software routine
d) interrupt synchronous routine
- Q.2 Which of the following file extension that is loaded in a microcontroller for executing any instruction?
a) .c b) .txt
c) .hex d) .doc
- Q.3 Which memory storage is widely used in Pcs and Embedded Systems?
a) EEPROM b) Flash memory
c) SRAM d) DRAM
- Q.4 What does ICE stand for?
a) in-circuit EPOM b) in-code emulation
c) in-circuit emulation d) in-code EPROM
- Q.5 If we say microcontroller is 8-bit then here 8-bit denotes size of:

- a) Data Bus b) ALU
c) Control Bus d) Address Bus
- Q.6 8051 series has how many 16 bit registers?
a) 2 b) 3
c) 1 d) 0
- Q.7 8051 microcontrollers are manufactured by which of the following companies?
a) Atmel b) Philips
c) Intel d) All of the mentioned
- Q.8 How many data lines are there in a 16*2 alphanumeric LCD?
a) 16 b) 8
c) 1 d) 0
- Q.9 Which of the following architecture is followed by general-purpose microprocessors?
a) Von Neumann architecture
b) Harvard architecture
c) None of the mentioned
d) All of the mentioned
- Q.10 Which forms the heart of the operating system?
a) Kernel b) Applications
c) Hardware d) Operating system

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is an AVR Microcontroller?
Q.12 What is a CAN Bus?
Q.13 What is SPI.

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- Q.14 What is pipelining?
Q.15 What is PIC Microcontroller?
Q.16 Why microcontrollers are not called general purpose computers?
Q.17 Define scan cycle.
Q.18 Explain Automotive Embedded System.
Q.19 What is UART?
Q.20 What is the Watchdog timer?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What are the characteristics of Embedded Systems?
Q.22 How the transfer C or ASM code in microcontrollers?
Q.23 What is interrupt latency? How can you reduce it?
Q.24 Explain what is the need for an infinite loop in embedded systems?
Q.25 Define the term Memory Management.
Q.26 Define the term Compiler.
Q.27 What are the advantages and disadvantages of Embedded Systems?
Q.28 Difference between AVR and PIC Microcontrollers
Q.29 What are the errors in an Embedded System?

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